



Autonomous Preservation Robots in Interplanetary Museum Vault scenario

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The report documents the solution to the space themed final project scenario, the Interplanetary Museum Vault (IMV): a subterranean Martian research outpost with some environmental constraints, in which robotic curators have to relocate artifacts to their appropriate destinations. The assignment is solved incrementally across the proposed five subproblems. The report follows the same structure, with one section per subproblem, documenting the scenario interpretations with consequent modelling choices, the planners used and why, and the results obtained.

1 Updates

This short revision paragraph addresses the three aspects reported in the evaluation: each problem section contains a subsection discussing the alternatives considered and why the adopted modeling choice was preferred; subsections about additional experiments and their results; each domain file now opens with a block comment explaining the key structural choices.

1.1 Problem 1

Several modeling decisions in Problem 1 were made considering the later problems, even though it only required a classical STRIPS formulation.

Items are organized under a common `item` supertype with `artifact` and `pod` as subtypes, allowing the predicates `at` and `carrying` to be used for both, avoiding duplicates.

The properties of artifacts are represented using two boolean predicates, `is-fragile` and `is-alpha`. This allows the two properties to be combined freely (e.g. `art-alpha-2` is both fragile and alpha). This choice was taken for simplicity: for example, using a subtype hierarchy for these properties would have been more complicated, as an artifact with both would require multiple inheritance, which is not handled cleanly by PDDL. Boolean predicates also allow any combination of properties to be represented without introducing additional types or actions.

Pods are modeled as physical objects that the curator retrieves and carries. This choice reflects the setting of the problem: the curator must travel to the Anti-Vibration Pods room, collect a pod, and then use it to transport the artifact. Representing the pod use with an abstract predicate such as a "stabilized" flag on the artifact would not have the same modeling effect. Making the pod a physical object also prepared the domain for Problem 2, where pods occupy the robot's carrying capacity like other objects. Pod capacity is intentionally not constrained in Problem 1, it only focuses on building the basic transport sequence (pickup, secure, travel, and unload). A pod can therefore hold any number of artifacts and the pickup action does not check if the pod is already occupied. The effect of this is visible in 2, where FF produces a plan that leaves a loaded pod at an intermediate location, a limitation of the domain model.

The `seismic-active` predicate and the distinction between `move-via-tunnel-to-beta` and `move-via-tunnel` are introduced even though seismic activity is never active to follow the environment model. When seismic activity becomes relevant in Problem 4, temporal constraints can therefore be added to the existing structure without redesigning the movement model.

1.2 Problem 2

Capacity is modeled using slot objects represented by `slot-of ?s ?r` and `slot-free ?s`. A numeric representation (such as `(load ?r)` with a fixed `(capacity ?r)` value) was con-

sidered because it would require only a single value per robot and therefore provide a more compact state representation. However, it would require support for `:numeric-fluents`. Explicit slot objects also provide more detailed information about the robot's load since the planner can reason about which item is occupying which slot. This is relevant to the optimal plan found, where the robot carries a pod in one slot and an artifact in the other at the same time.

Robot capabilities are represented with Boolean predicates, `can-fly` and `can-carry-pod`, on a single `robot` type. Separate `curator` and `drone` subtypes were not used for the same reason as the artifacts in Problem 1. As a consequence, actions such as `pickup-pod` and `fly` can remain generic over `?r - robot`, with the required capability specified as a precondition. This also makes the model easier to extend: a future robot with both capabilities could be created assigning both predicates without introducing a new type or a separate set of actions. Moreover, using separate robot subtypes would also make the capacity model more complicated, since slots belong to individual robot instances independently of their capabilities.

Slots are associated with individual robots with the `slot-of ?s ?r` predicate. Each robot therefore manages its own resources independently, and the planner only needs to choose which slot belonging to that robot should be used.

Finally, Hall β is identified using the predicate `is-hall-beta ?l`. It is an ordinary location but has special connectivity represented by the tunnel predicates from Problem 1. Representing it as a predicate allows the same location object to be handled consistently by both the existing ground travel actions and the new flight actions. This avoids introducing a location hierarchy still allowing the planner to distinguish Hall β when needed.

1.3 Problem 3

The five `t-handle-artifact` subtasks of `m-mission` are totally ordered (`task0 < task1 < ... < task5`). The total ordering was therefore chosen to provide a direct and unambiguous representation of the mission, at the cost of the optimization opportunities that are later visible and discussed in the results.

`m-mission` takes the robot as a parameter `?c`, with the precondition (`can-carry-pod ?c`). In the current problem instance, only `curator1` satisfies this capability, so the parameter is forced on that robot, but the free variable formulation is present so that the method would be valid if a future problem instance included more than one robot capable of carrying pods.

The four possible combinations of the artifact properties (alpha and/or fragile) are represented by four separate and mutually exclusive `t-handle-artifact`. HDDL methods don't have a direct if/then construct, and there are only four possible combinations. Explicitly listing them therefore keeps each method as a simple and readable sequence of subtasks, following the observation in section 4.2 that the HTN representation is easier to read than the corresponding flat PDDL formulations. The same principle is used for `t-transport-fragile`, which has two separate methods for the cases with and without cooling. The additional `cool-in-pod` action is duplicated across the two methods. This was considered simpler than introducing another layer of task decomposition by creating a common sequence with an optional cooling task.

The recursion in `t-goto` uses one tunnel hop or one flight per recursive call, with `m-goto-arrived` as the base case. This structure exploits the map's radius 1 star topology: every location is at most one intermediate hop away from the Maintenance Tunnel. Therefore, ground travel requires at most two recursive calls. An alternative would have been to define a separate non recursive method for every concrete (from, to) location pair

but this approach was rejected because it could require many more methods and would require additional ones with a map extension.

The seismic condition is included in the method definitions `m-goto-ground-step-beta`, `m-goto-fly-to-beta`, and `m-goto-fly-from-beta`, even if the corresponding primitive actions already enforce the same condition. The duplication is intentional: the HDDL method preconditions are checked before the method is decomposed, so an invalid branch can be rejected immediately rather than being expanded until the primitive action is reached. The additional condition therefore improves search efficiency without changing the set of valid plans.

1.4 Problem 4

Manipulation actions enforce mutual exclusion with movement by reusing (`over all (robot-at ?r ?l)`). Reusing `robot-at` was preferred because it prevents movement during manipulation without requiring additional predicates with the only purpose of controlling concurrency.

It's also relevant to notice that the reformulation required by OPTIC and POPF (section 5.3) introduces a new modeling cost. For example, `pending-cool` and `cooled` now represent complementary aspects of the same condition and must be initialized and maintained separately. `pending-cool` is initially true for each alpha artifact and is set to false at the start of `cool` or `cool-in-pod`. In Problem 2, this information is instead represented directly using (`not (cooled ?a)`). This is therefore a modeling cost introduced by the need to work around the planners' limited support for negative preconditions.

1.5 Problem 5

The changes already discussed in section 6.2 also required widening the parameter list of `cool` and `cool-in-pod` from (`?r ?a ?s`) to (`?r ?a ?s ?l - location`). The location parameter is necessary because the `over all` condition must refer to a variable that is explicitly bound as an action parameter. The parameter is used only by the PDDL planner and executor to verify the location invariant and doesn't affect the corresponding ROS2 action-performer nodes, which do not read or use `?l` and execute the cooling action as before. This illustrates the separation between the PDDL planning model and the ROS2 execution layer: the planning model can be extended with additional information needed for temporal reasoning without requiring a corresponding change to the action-performer implementation.

1.6 Additional Problems

In addition to the alternative map topology experiment previously discussed in the Bonus section 4, three more variations were tested using the Problem 2 and Problem 4 domains. These experiments were designed to test scalability and the effects of different environmental constraints. Screens of the plan runs are in the Appendix [16, 17] .

1.6.1 Scaled Artifact Count

This problem doubles the number of artifacts from 5 to 10 and keeps the map, robots, capacities, and pods unchanged. The distribution was chosen so that each hall contains two fragile artifacts. This was done to test if a larger number of fragile artifacts would require the use of both available pods and change the planning schedule.

The satisficing planners solved the larger instance quickly: LAMA-first found a 51 step plan in 0.008 s expanding 192 states, FF found a 47 step plan evaluating 122 states to depth 3. In contrast, Downward with A*/LM-cut, couldn't terminate within the 5 minute and 4 GB limits given. Before the timeout, it had completely explored the $f = 37$ search frontier and evaluated more than 2 million states. Therefore, the optimal plan has a cost of at least 38, although its exact cost could not be determined. Memory usage was only 121 MB, showing that the limiting factor was search time, as intuitively guessed during the first run that led to setting run constraints. This is a significant change from the original 5 artifact instance, where the cost optimal search terminated in only 0.17 s. Therefore, simply doubling the number of artifacts was enough to make optimal search fail to terminate in an acceptable time period, showing a scalability limitation.

The two completed plans also provide further evidence of the pod capacity limitation discussed in the report. Neither LAMA-first nor FF ever uses the second pod. Instead, both plans exploit the fact that the current model does not restrict the number of artifacts that can be stored in a pod. At both Hall α and Hall β , the plans load the two fragile artifacts into the same pod before unloading either of them.

Metric	LAMA-first	FF	FD (A*/LM-cut)
Search strategy	Greedy BFS + landmarks	Enforced hill-climbing	A* (optimal)
Heuristic	FF + landmark heuristic	Relaxed-plan (h_{FF})	LM-cut
Optimality guarantee	No	No	Yes
Plan length	51	47	≥ 38 (unsolved)
States expanded	192	~ 122	64,963
States evaluated	193	122	456,631
Search time	0.0077 s	< 0.01 s	> 300 s (limit reached)
VAL validation	Valid	Valid	-

Table 1: Comparison of planners on the scaled up version of Problem 2

1.6.2 Pod Capacity

Sections 2.3, 3.3, and 1.6.1 identified a limitation that had been present since Problem 1: **secure-in-pod** does not check if the pod already contains an artifact. As a result, a single pod can hold an unlimited number of fragile artifacts before any of them is unloaded. This is different from the robot slot capacity introduced with Problem 2 (section 3.2), which limits how many objects a robot can carry but does not limit the number of artifacts inside a carried pod. To evaluate the effect of fixing this specific limitation, a new domain was created from the 10 artifact scaleup problem. A binary **pod-empty** predicate was added: **secure-in-pod** now requires the selected pod to be empty and clears **pod-empty** when the artifact is loaded, **unload-from-pod** restores it.

Both satisficing planners found valid plans, but the plans, as expected, became longer. LAMA-first increased from 51 to 54 steps, while FF increased from 47 to 58 steps. The number of evaluated states also increased, approximately doubling for FF and growing moderately for LAMA-first. FD/A*-LM-cut again did not terminate within the given 5-minute/4GB limits. However, it progressed further than on the unconstrained instance: this run fully closed the $f = 38$ frontier at $t = 192$ s, proving a cost lower bound of ≥ 38 . This matches the lower bound reached by the unconstrained instance, where the search had only fully closed the $f = 37$ frontier within a comparable amount of time. An additional run with no enforced time or memory limit was manually interrupted after

approximately 25 minutes, with a peak memory usage of 1.75 GB. Before being stopped while searching the $f = 39$ frontier, it had increased the proven lower bound to ≥ 39 .

Metric	LAMA-first	FF	FD (A*/LM-cut)
Plan length (steps)	54	58	≥ 39 (unsolved)
Δ length	+3	+11	$\geq 38 \rightarrow \geq 39$
States expanded	252	~ 244 (max depth 3)	1,368,203
States evaluated	253	244	9,107,495
Peak memory	12.9 MB	-	1.75 GB
Search time	0.0079 s	< 0.01 s	> 1471 s (interrupted)
VAL validation	Valid	Valid	-

Table 2: Results and comparison of planners on the pod capacity scaleup problem with the scaleup problem

Reading the completed plans it can be noted that instead of intuitively carrying both pods and securing two fragile artifacts at the same time, both LAMA-first and FF use the same sequential and reuse strategy saw: one pod is repeatedly loaded, unloaded, and reused for the fragile artifacts. LAMA-first picks up `pod1` at the beginning and keeps it in `slot-c1a` throughout the entire plan, reusing it for all four fragile artifacts. FF follows the same strategy with `pod2`. Neither planner ever uses the second physical pod. Therefore, the pattern observed in Problems 1, 2 and in the scaleup problem persists even after the possibility of storing multiple artifacts in one pod has been removed.

The FF plan exposes an inefficiency caused by the capacity constraint. In Hall α , it secures and cools `art-alpha-2`, then immediately unloads it again (steps 12–14) to free the pod for `art-alpha-3`. Much later, at step 42, `art-alpha-2` has to be secured in the same pod again before it can be delivered. This creates a redundant unload/reload cycle. LAMA-first avoids this by processing the two fragile artifacts in each hall consecutively before leaving the hall.

In conclusion, together with the previous results, this shows that the pod constraint increases the plan length, the single pod reuse strategy remains unchanged, and the 10 artifact instance remains difficult for optimal search.

1.6.3 Wider Seismic Window

The second experiment changes the timing of the seismic activity. The original two windows are replaced by a single larger window [8,28]. The purpose of this experiment is to investigate whether the placement and structure of the seismic period can be more relevant to the planning than its total duration.

OPTIC found a plan with a duration of 41.001. This is only slightly longer than the non seismic baseline of 40.007 minutes and is shorter than the 45.001 minute solution obtained with the original two window configuration. Therefore, the longer window doesn't necessarily produce a longer plan. The plan explains this result. Because there is only one continuous closure, the robots can organise their Hall β operations around a single reopening. The drone crosses Hall β at $t = 0$, before the seismic window begins. The curator then starts its crossing at $t = 25$, so that the 3 minute crossing ends exactly when the window closes at $t = 28$. The robots can therefore group their Hall β operations into a single coordinated pass.

POPF found a 54.009 plan. In this solution, the trip to carry the pod and the delivery of the non fragile alpha artifact are treated as separate trips instead of being combined. This

creates additional tunnel travel. The difference between the OPTIC and POPF solutions is therefore larger than in the original seismic instance. In this case, the difference can be interpreted as a consequence of the different search behaviour of the two planners on this particular temporal problem.

Planner	Two windows (original)	Wide window (new)	Actions (original / new)
OPTIC	45.001	41.001	33 / 32
POPF	54.009	54.009	32 / 32

Table 3: Comparison of results with a wide seismic window ($t \in [8, 28]$) vs. the two windows ($t \in [5, 15]$, $t \in [25, 32]$)

2 Problem 1

2.1 Problem Description

A single robotic curator, initially at the vault’s entrance, must relocate artifacts spread across the vault to their destinations. Three artifact characteristics are modelled, and can be combined on the same object:

- **Fragile:** artifacts must be secured inside an anti-vibration pod before being moved.
- **Alpha:** artifacts must be cooled and delivered to the Cryo-Chamber.
- **Core samples:** must be delivered to the Stasis-Lab.

The vault map (Figure 1 of the assignment) is a star topology: the Maintenance Tunnel is the single central corridor, and all other rooms connect to it. Crossing into the tunnel from any room requires the curator to have its sealing mode active (**sealed**). Hall β is only reachable while no seismic window is active (**seismic-active**). Since Problem 1 is non temporal, this predicate is left false throughout the initial state and the plan. It is assumed the whole operation happens within a single non seismic window; the predicate becomes time varying when durative actions are introduced in Problem 4.

The anti-vibration pods were modelled as physical objects the curator carries. Pods start on the floor in the Anti-Vibration Pods room and are picked up like other items. Fragile artifacts can never be picked up directly, the **pickup-artifact** action explicitly excludes them, so the only way to relocate one is the **secure-in-pod** action, whose precondition requires the curator to already be carrying a pod. This enforces the wanted sequence of actions: retrieve a pod first, travel to the artifact, load it in the pod, travel to the destination, and finally unload it. A **stabilized** flag records that an artifact has been put in a pod and is checked as part of the goal.

2.2 Planner selection and Solving

The domain is expressible in plain STRIPS with typing and negative preconditions, no numeric fluents or temporal constructs are needed for Problem 1. Any classical planner in the **planutils** toolbox is applicable. Three configurations were chosen and ran for comparison:

- **LAMA-first:** a satisficing planner based on Fast Downward, uses a landmark guided greedy best first search.

- **FF**: a satisficing planner using enforced hill climbing guided by the relaxed plan heuristic h_{FF} .
- **Fast Downward, A*/LM-cut** (`--search "astar(lmcut())"`): an optimal search using the admissible LM-cut heuristic.

The last one was chosen specifically to obtain a lower bound on the plan cost to compare the other two non optimal planners' solutions and solving time. All three plans were validated with VAL against the domain and problem files using the command:

```
planutils run val -- Validate domain.pddl problem.pddl sas_filename.plan.
```

2.3 Results

Metric	LAMA-first	FF	FD (A*/LM-cut)
Search strategy	Greedy BFS + landmarks	Enforced hill-climbing	A* (optimal)
Heuristic	FF + landmark heuristic	Relaxed-plan (h_{FF})	LM-cut (admissible)
Optimality guarantee	No	No	Yes
Plan length (steps)	35	36	34
Plan cost (unit cost)	35	36	34
States expanded	79	~65 (max depth 3)	185,969
States evaluated	80	65	438,377
Search time	0.0016 s	<0.01 s	14.97 s
VAL validation	Valid	Valid	Valid

Table 4: Comparison of planners on Problem 1

Both satisficing planners found valid plans close to the lower bound optimum of 34 established by the Fast Downward's A*/LM-cut: LAMA-first was one step above optimal (35), FF two steps above (36). The inspection of the generated plans shows why: LAMA-first fetches each of the two antivibration pods only once and uses it continuously until it is no longer needed, whereas FF's plan parks a pod mid-transport with an artifact still sealed inside it and later re-collects it to load a second artifact. In this domain it was legal because pod capacity is unbounded and `putdown-pod` does not need to check `in-pod`, but is two actions less efficient.

The cost of optimality can be observed: A* expanded 185969 states and evaluated 438377 over roughly 15 seconds, versus the satisficing planners' 79 and 65 expanded states in a few milliseconds. It's a demonstration of the optimality-speed trade off between incomplete guided search and exhaustive optimal search, even on a small planning instance.

A limitation of the Problem 1 domain was exposed by FF's plan: `putdown-pod` does not clear `in-pod` contents and there is no pod-capacity constraint, so a loaded pod can be parked during transport, or two artifacts could be double loaded in the same pod simultaneously.

3 Problem 2

3.1 Problem Description

Problem 2 keeps the same initial conditions and goals of Problem 1. It extends the agent side, adding new capabilities to the curators and a maximum capacity constraint.

Depending on the subtype of `robot`, capabilities of the curator are expressed as predicates (`can-fly`, `can-carry-pod`):

- **curator1**: retains Problem 1’s tunnel travel and sealing requirement, and is the only agent with **can-carry-pod**.
- **drone1**: bypasses the Maintenance Tunnel sealing with **fly** action and lacks **can-carry-pod**, not being able to move fragile artifacts.

The maximum capacity is modelled with a fixed pool of *slot* objects per robot (**slot-of ?s ?r**), each independently free or holding one item. **curator1** has two slots, **drone1** has one. This keeps the domain within plain **:strips :typing :negative-preconditions**, so that it remains solvable by the previous classical planners without resorting to numeric planners. A relevant consequence of the two slots curator is that it can carry a pod in one slot and a common artifact in the other one, simultaneously.

3.2 Results

The optimal plan (cost 26) is the most informative of the three. It makes almost no use of the drone, only for the core sample, whose handling has no special requirements. It relies only on the robotic curator, exploiting the two slot capacity to combine two deliveries per room visit. Neither LAMA-first (31 steps) nor FF (28 steps) use this strategy. They both spread tasks evenly across the two agents, which intuitively looks like better use of multi-agent but costs more steps overall.

It can be concluded that between the two extensions introduced, the *capacity* was the more impactful factor for minimizing plan length, since a single well loaded curator run outperforms the parallel tasks plans found by the satisficing planners.

Metric	LAMA-first	FF	FD (A*/LM-cut)
Optimality guarantee	No	No	Yes
Plan length / cost	31	28	26
States expanded	111	~58 (max depth 2)	38,402
States evaluated	112	58	194,577
Search time	0.0035 s	<0.01 s	8.04 s
VAL validation	Valid	Valid	Valid

Table 5: Comparison of planners on Problem 2

4 Problem 3

4.1 Problem Description

Problem 3 keeps the previous scenario, primitive actions, initial conditions, and goals, encoding the objective as a task network. All primitive actions were maintained from Problem 2’s domain. A **:task/:method** decomposition layer was added on top.

4.2 Task hierarchy

Figure 1 shows the decomposition of the top level tasks. The case of **art-alpha-2**, both alpha and fragile, is expanded completely down to the primitive actions because it is the only case that requires all the mechanisms of the hierarchy. The other three **t-handle-artifact** methods have the same general structure, but use either **t-transport-plain** instead of **t-transport-fragile**, omit the cooling action, or both, as summarised in Table 6. Figure 2 shows the recursive structure of **t-goto** separately.

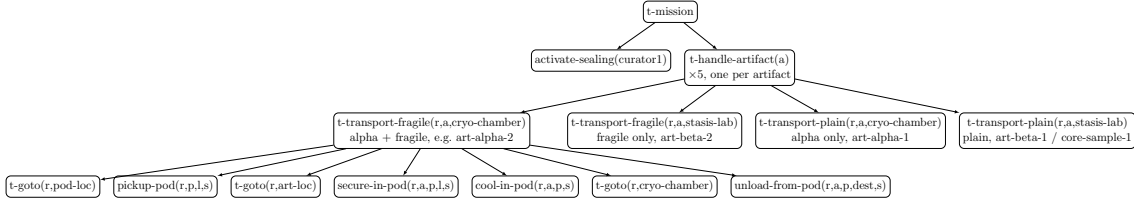


Figure 1: Problem 3 task hierarchy. **t-mission** down to primitive actions, with the alpha+fragile branch (**art-alpha-2**) expanded.

Method	Route	Cooling step	Destination
m-handle-alpha-fragile	pod	cool-in-pod	cryo-chamber
m-handle-fragile-only	pod	-	stasis-lab
m-handle-alpha-only	direct	cool	cryo-chamber
m-handle-plain	direct	-	stasis-lab

Table 6: Summary of the four **t-handle-artifact** methods

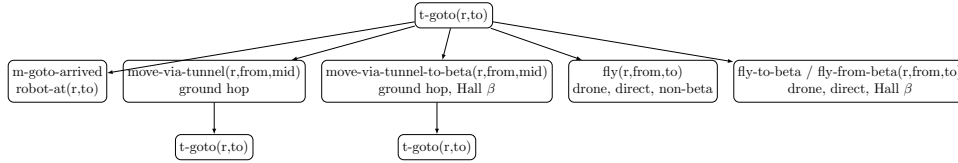


Figure 2: **t-goto** five methods

The top level method, **m-mission**, is specific to the problem instance. It explicitly defines the five named artifacts as ordered subtasks: the top level method describes the specific mission to be completed, while the tasks below **t-handle-artifact** remain generic and reusable. The curator assignment is left open to the planner whenever Problem 2 allowed multiple possible assignments. The methods of **t-handle-artifact** only require the selected robot to satisfy **can-carry-pod** when the artifact is fragile. Therefore, the HTN encoding does not unnecessarily restrict the set of valid robot assignments defined in Problem 2. Movement is represented by the recursive task **t-goto**. Its structure exploits the star shaped topology of the map (radius 1 around the Maintenance Tunnel), in which every location is at most one intermediate hop away from the Maintenance Tunnel. The task includes a base case for when the robot is already at the destination and recursive methods that perform one movement step toward the destination before calling **t-goto** again. Separate methods are defined for the available travel modes.

For ground movement, the recursion terminates after at most two levels because the required route can contain at most one intermediate location. Flight doesn't need recursion because the drone can always reach its destination in a single action.

In general, the HTN approach led to a modular and clearer structure, providing an advantage in the code writing and interpretation compared to the previous PDDL problems.

4.3 Planner selection and Solving

pandaPIparser/panda is the only HTN planner in the **planutils** list. Panda was run with its default configuration: a greedy best-first progression search guided by the **hRC2** heuristic, stopped at the first solution found. So the obtained plan is a non-optimal

configuration.

4.4 Results

Metric	PANDA
Search strategy	Greedy HTN progression, stops at first solution
Heuristic	hRC2 (relaxed composition, FF-based)
Optimality guarantee	No
Primitive actions in plan	37
of which movement	22
of which artifact actions	14
Search time	0.005 s
Nodes generated	745
Nodes visited	216

Table 7: Results for Problem 3 (HTN formulation) using `panda`

The plan found is valid: the final state has all five artifacts reaching the corresponding destinations with the required `stabilized/cooled` flags set. Since standard `VAL` expects a flat `:goal` to check, the validation for this problem was done by manually inspecting the final state reached.

Panda’s 37 actions are higher than all three previous solutions. Two decomposition choices account for the gap. First, the plan retrieves *two* separate pods (`pod1` for `art-beta-2` and `pod2` for `art-alpha-2`), each requiring its own round trip to the Anti-Vibration Pods room, rather than reusing a single pod sequentially the way Problem 2’s optimal plan did. Second, it never combines two deliveries into a single room visit the way Problem 2’s optimal plan did at Hall β , collecting the pod and a loose artifact together using both of the curator’s slots). Here, `art-beta-1` and `art-beta-2` are each fetched with a separate visit to Hall β . Both are consequences of `t-transport-fragile`’s methods choosing a pod and a room visit locally, without any mechanism to notice that a pod or a trip already in progress could serve a second artifact transport. This is relevant because it’s the kind of optimization a greedy, first-solution HTN search has no incentive to find. In conclusion, we can observe that an HTN’s task structure encodes how to decompose a job, but the opportunities that shorten the plan across separate task instances are only found if the search are explicitly built to look for them.

Panda reports the task decomposition graph contains 5 cyclic strongly connected components, each of size 1. These are self loops coming from `t-goto`’s recursive definition. Panda also confirms the instance is totally ordered, consistent with using `:ordered-subtasks` in the domain.

5 Problem 4

5.1 Problem Description

Problem 4 converts Problem 2’s domain to durative actions: every action has now a chosen fixed duration (Table 8), and the domain allows concurrent execution.

As specified in the assignment the following action overlaps were avoided: a curator picking up two artifacts at once, and a drone picking something up while simultaneously flying off. To obtain it: Movement actions delete `robot-at at start` and only restore it at

Action	Duration (min)	Description
move-via-tunnel / -to-beta	3	Ground travel in the tunnel
fly / fly-to-beta / fly-from-beta	2	Fly is faster than ground travel
activate-sealing / deactivate-sealing	1	Fast mode switch
pickup-* / putdown-*	1	Simple grab/release
secure-in-pod / unload-from-pod	2	Careful handling of a fragile items
cool / cool-in-pod	3	Slower than mechanical actions

Table 8: Actions' duration chosen for Problem 4.

end, so for the whole duration of a move the robot is in transit, having no location, while every manipulation action (`pickup-*`, `putdown-*`, `secure-in-pod`, `unload-from-pod`) requires (`over all (robot-at ?r ?l)`), pinning the robot in place for its entire duration. Because a move deletes the very fact a manipulation needs to persist, the two can never be scheduled to overlap on the same robot. In the same way it is prevented `deactivate-sealing` during tunnel crossing, since `move-via-tunnel/-to-beta` requires (`over all (sealed ?r)`).

The domain allows *parallelism between the curators*: the different robots never share a predicate instance, so `curator1` and `drone1` act independently.

Two *manipulations in different slots* on the same robot (e.g. picking up a pod in one hand while picking up a loose artifact in the other) are unconstrained since they consume different `slot-free` facts.

Cooling (`cool/cool-in-pod`) is tied only to (`over all (holding ...)`), not to `robot-at`, modelling cooling units that keeps running while the robot continues to travel.

5.2 Seismic events

Seismic activity, always set to false in the previous Problems 1–3, has now real temporal meaning using `:timed-initial-literals`. Two quake windows were simulated ($t \in [5, 15]$ and $t \in [25, 32]$ minutes). To do implement it, they are declared in `:init` as scheduled facts independent of any action a planner might choose to take.

A separate, non-seismic variant of the same domain and problem was also produced (`seismic-clear` asserted once and never modified) specifically to simulate and measure a comparison against Problems 1–3 alongside the temporally-constrained version.

5.3 Planner selection and Solving

From the `planutils` catalog, OPTIC, POPF (COLIN based), and TFD (Temporal Fast Downward, Fast-Downward based) were run successfully on the non-seismic domain. TFD's translator rejects `:timed-initial-literals`, so it could only be run for the non-seismic simulation; OPTIC and POPF both accepted the seismic variant, though as discussed below POPF's seismic result should be treated with caution.

A first version of this domain reused Problem 2's negative preconditions. OPTIC and POPF, the chosen planners, rejected it. As also mentioned during the course, despite accepting `:negative-preconditions`, neither actually implemented negated literals in action *preconditions*, so a refactoring was needed. The fix was to modify every negative precondition into a complementary positive predicate: (`not (sealed ?r)`) $\rightarrow$ (`unsealed ?r`); (`not (is-fragile ?a)`) $\rightarrow$ (`not-fragile ?a`); (`not (cooled ?a)`) $\rightarrow$ (`pending-cool ?a`); (`not (is-hall-beta ?l)`) $\rightarrow$ (`not-hall-beta ?l`); and (`not (seismic-active)`) $\rightarrow$ (`seismic-clear`).

5.4 Results

Planner	Non-seismic	Seismic	Actions (non-seismic / seismic)
OPTIC	40.007*	45.001*	29 / 33
TFD	40.161	-	32 / -
POPF	54.009	54.009 [†]	32 / 32

Table 9: Results for Problem 4

*OPTIC found this valid plan and continued its search for a better one, as instructed by the declared `(:metric minimize (total-time))`. Both runs timeout the planner before proving optimality, so the reported ones are the best solutions found 8, 10. In both cases the cost is already at or below OPTIC’s optimistic admissible cost estimate, so a strong evidence of near optimality.

[†]POPF produced identical plans for the seismic problem and the non-seismic one. It had a Hall β crossing at $t = 25.006$ that falls inside the second quake window ($t \in [25, 32]$). If `:timed-initial-literals` were being correctly enforced this would be an invalid plan, so this result is reported for completeness but is not considered a correct seismic scheduling. More explanations below.

The non-seismic results cross validate between two of the three planners. OPTIC’s captured solution (40.007) and TFD’s rescheduled solution (40.161) agree both choosing on the same strategy: the drone handles the two artifacts with no special requirements (`art-beta-1`, `core-sample-1`) plus `art-alpha-1`, while the curator retrieves a single pod once and reuses it sequentially for both fragile artifacts, similarly to the pod reuse strategy Fast Downward’s optimal search found for the Problem 2.

POPF’s result (54.009, 32 actions) is distant from the other two solutions. It makes the drone deliver only `art-beta-1` and `core-sample-1`, leaving `curator1` to pick up, cool, and deliver `art-alpha-1` itself with plain `pickup-artifact/cool/putdown-artifact` sequence before it goes to fetch a pod for the two fragile items. This leaves the drone in idle after the first two deliveries and adds an extra trip and manipulation sequence to the curator’s path. Since POPF performs a single greedy forward search rather than an optimization loop, it never gets the opportunity to reconsider this plan once made. This shows how much a single early scheduling decision (which agent takes `art-alpha-1`) can impact when a planner commits to it without revisiting.

Comparing the seismic and non-seismic OPTIC results directly quantifies the cost of the quake constraint: duration rises from 40.007 to 45.001, due to the plan needing to schedule the Hall β visits around the two seismic windows rather than freely.

POPF’s seismic result is excluded from the comparison, for the reason reported in the table footnote. A detailed comparison of POPF’s non-seismic and seismic plans shows that the two solutions are identical in both their action sequence and timestamps. In particular, the plan still schedules the curator’s crossing of Hall β at $t = 25.006$, within the second earthquake interval, $[25, 32]$.

This behaviour seems to not be caused by POPF ignoring the `:timed-initial-literals` (TILs) in the problem. It reports 93 literals for the non-seismic problem and 95 for the seismic version, so the two additional TIL facts are successfully parsed and added.

To further test the issue, the `seismic-clear` condition of the Hall β crossing actions was changed from an `over all` invariant to a `at start` precondition. However, POPF

returned the same plan, including the same Hall β crossing at the same invalid timestamp. This result shows that the problem is not specifically related to the handling of **over all** invariants. In conclusion, the results suggest a planner’s limitation. POPF recognises the the TIL literals but doesn’t correctly incorporate the constraints to generate and validate the final plan.

The same reformulation was also tested with OPTIC as a control. In contrast to POPF, OPTIC correctly handled both formulations. With both the **over all** and **at start** versions, the Hall β crossing was scheduled immediately after the end of the second earthquake interval, at $t = 32.000$ and $t = 32.001$, respectively. The resulting plans had almost identical duration (45.001 and 45.002) and required a comparable number of evaluated states (2144 and 2431). This demonstrates that changing the condition from an invariant to a start precondition does not introduce ambiguity or make the problem inherently difficult for a temporal planner.

The comparison with OPTIC therefore further supports the conclusion that the problem is specific to POPF. Using the same domain and problem encoding, OPTIC was able to produce a different plan that correctly considered the earthquake intervals, regardless of whether seismic-clear is expressed as an over all invariant or an at start precondition.

The parallelism and scheduling are directly verifiable in the seismic OPTIC plan’s timestamps. *Cooling overlapping movement:* **curator1** begins **cool-in-pod** and **move-via-tunnel** simultaneously at $t = 19.005$, both finishing at $t = 22.005$, so the fragile, alpha-typed artifact is cooled while the curator is already travelling toward the cryo-chamber with it. (Overlap shown in Figure 3 or 10).

Scheduling around both quake windows: **drone1** visits Hall β once, from $t = 20$ to $t = 25$, in the interval between the first window (ends $t = 15$) and the second (starts $t = 25$); **curator1** visits Hall β once, starting its tunnel crossing at $t = 32.000$, the instant the second window closes and **seismic-clear** becomes true again.

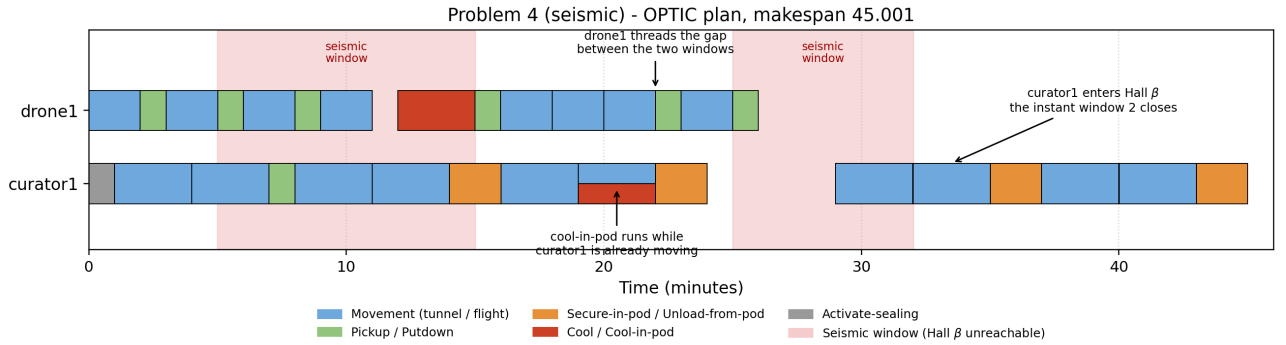


Figure 3: Gantt chart of the seismic OPTIC plan (Table 9). In the picture one row per agent, coloured by action category. The red bands highlight the two seismic windows. As discussed above: the cooling/movement overlap on **curator1** and each agent’s Hall β visit landing in the gap between windows

The termination on the seismic OPTIC search, instead of a clean stop is a direct consequence of the domain’s capacity model. Named robot slot objects (**slot-c1a**, **slot-c1b**) mean every manipulation action is grounded once per slot, per pod, per location. This means we need a trade-off: the richer the model used, the more expensive is to search over concurrent scheduling.

6 Problem 5

6.1 Problem Description

Problem 5 deploys the durative domain and problem of Problem 4, in the non-seismic variant, inside PlanSys2. It uses one ROS2 fake action performer node per PDDL durative action, following the PlanSys2 "first planning package" tutorial pattern, in which each performer subclasses `ActionExecutorClient`, reports progress via `send_feedback()`, and calls `finish()` on completion.

6.2 PlanSys2 Adaptation and Considerations

Several issues encountered during the implementation were related specifically to the PlanSys2 toolchain, as a consequence, the PDDL model developed in Problem 4 was adapted. Below are described the changes and reason behind them.

Type hierarchy: PlanSys2's PDDL parser caused a segmentation fault in `plansys2_node` when loading the type hierarchy used in Problem 4, where both `artifact` and `pod` were subtypes of `item`. The problem was still present even when no action performer nodes were running, indicating that the failure occurred during domain parsing. To solve this issue, the type hierarchy was flattened by removing the common supertype. In addition, predicates that previously accepted the general `item` type were replaced by type specific predicates: `at` became `at-artifact` and `at-pod`, `holding` became `holding-artifact` and `holding-pod`.

Negative preconditions: PlanSys2's default planner, POPF, rejects the negative preconditions used in the domain. This limitation had already been encountered in Problem 4 using POPF. Therefore, the same solution was reused: negative preconditions were converted to equivalent positive predicates.

Durative actions required for execution: a version of the domain using classical (`:action`) definitions was accepted by the planner and it successfully generated a valid plan. However, the execution fails with the error *"Requesting a non-existing durative action"*. This showed that regardless of how the plan was generated, PlanSys2's executor expects every action to be defined as a `:durative-action`. So all actions in the executable domain had to be represented as durative actions.

ROS2 node naming constraint: `ActionExecutorClient` uses the PDDL action name as the name of its corresponding ROS2 node. However, ROS2 node names must contain only alphanumeric characters and underscores. As a result, actions with hyphens in their names, such as `move-via-tunnel`, caused their performer nodes to terminate with an `InvalidNodeNameError`. All action identifiers were therefore renamed using underscores (e.g. `move_via_tunnel`).

Race condition during cooling: The original `cool` and `cool_in_pod` actions did not include a condition requiring the robot to remain at its current location. As a consequence, POPF was allowed to schedule cooling concurrently with a movement action involving the same robot. This behaviour is valid similar to the action overlap used in the seismic OPTIC plan described in Problem 4. However, during execution, independently timed ROS nodes can't guarantee the synchronization assumed by the temporal plan, leading to a plan failure. To avoid this, the condition `(over all (robot-at ?r ?l))` was added to both cooling actions. This forced cooling and movement actions involving the same robot to execute sequentially. Although this removes the action parallelism for the planner, it provides a way to perform the execution with ROS2 nodes.

6.3 Planner and Execution

POPF is PlanSys2's default plan solver, as used in Problem 4. `plansys2_terminal` has two ways to load a `pddl` problem: `incremental set instance/set predicate/set goal` commands, typed or sourced from a file, or a standard PDDL problem file loaded as a `problem.file` ROS parameter. Both the approaches were experimented during the development but the first was found often unreliable at this problem's scale (e.g. during startup the first command could silently be dropped). The problem file approach was used to avoid issues. In the README file are described in detail the commands used.

6.4 Results

The plan found consists of 34 actions with a makespan of 43.007, using the time units from Problem 4 duration table 8. With `plansys2_terminal` run command every action reported progress to 100% and the executor logged **Plan Succeeded**.

The strategy found is structurally similar to Problem 4 non-seismic solutions. The drone independently handles `art-beta-1`, `core-sample-1`, and `art-alpha-1`, while the curator retrieves a single pod and reuses it sequentially for both fragile artifacts. The solution duration (43.007) sits between Problem 4 OPTIC/TFD results (40, Table 9) and POPF result (54.009). It is not directly comparable to either, since the `robot-at` removes the cooling/movement overlap used in Problem 4, but the slightly longer duration is logically coherent with the change.

7 Problem 5 - Bonus

To further experiment with Problem 5 implementation, a change in the map topology was tested. The same `domain.pddl` and all 15 fake action-performer nodes were reused with a new problem instance changing only `problem.pddl`.

7.1 Map topology

The map variant renames every object and location to a Martian theming setting inspired by the book and movie "The Martian". The map is reshaped to reflect the geography of the described environment: the Hab is a single inflatable dome, so its interior (`hab-corridor` connecting to the Hydroponics Bay, Med Bay, Sample Locker, and Workshop) remains a star, structurally identical to the original IMV topology. Surface excursions outside the Hab to recover the Pathfinder probe are modelled as journeys broken into waypoint camps. Reaching the Pathfinder site takes three tunnel crossings from the airlock (`airlock` → `waypoint-1` → `waypoint-2` → `pathfinder-site`), with the final step gated by dust-storm windows, as happened for Hall β with seismic events.

7.2 Planner and Execution

The hybrid shape isolates the cost increase in the one route that is supposed to be expensive. It's a way to test if the planner shifts work to the drone when ground travel becomes expensive.

7.3 Results

For the same goal of collecting five artifacts, the plan duration increased from 43.007 to 75.015. The distribution of actions between the rover and the drone also changed

IMV concept	Martian Hab equivalent
Entrance	Airlock
Maintenance Tunnel	Hab Corridor
Hall α (temperature-sensitive)	Hydroponics Bay
Cryo-Chamber	Med Bay
Stasis-Lab	Sample Locker
Anti-Vibration Pods room	Workshop
Hall β	Pathfinder Site
-	Waypoint 1 $\rightarrow$ Waypoint 2
-	Solar Farm (short EVA branch)
curator1 (ground, carries pods)	rover1
drone1 (aerial)	recon_drone1
Anti-vibration pod	Transport crate
Artifact: fragile (art-beta)	Soil sample
Artifact: temperature sensitive (art-alpha)	Botany specimen
Artifact: core sample	Geological core sample
Seismic activity	Dust storm

Table 10: Mapping of the topology redesign

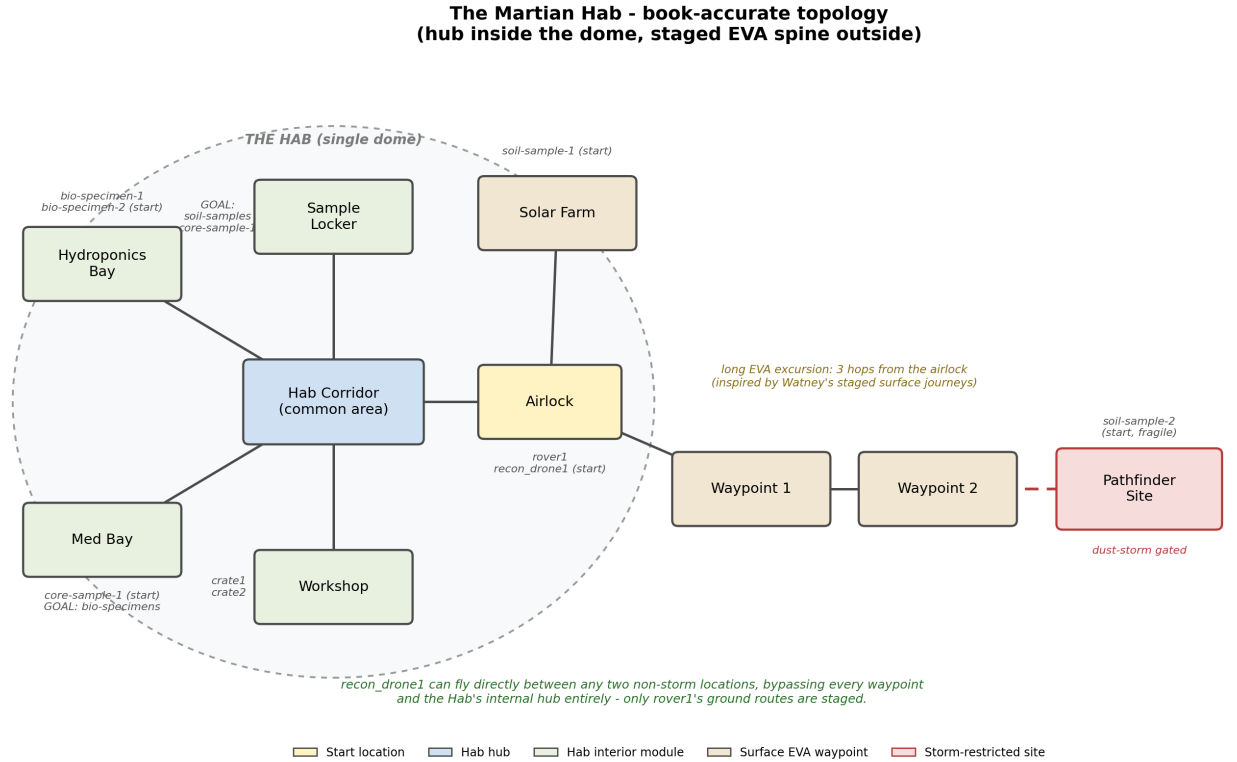


Figure 4: The Martian inspired map topology

significantly, from an even 17/17 split to 29 actions for the rover and 12 for the drone.

However, this difference did not indicate a major change in the planner's strategy. Only the rover can transport a pod, meaning that the two fragile artifacts, **bio-specimen-2** and **soil-sample-2**, had to be assigned to the rover in both topologies. The main difference is that, in the new map, reaching **soil-sample-2** requires travelling through a longer

Metric	Original	Martian
Plan length (actions)	34	41
Duration	43.007	75.015
Action split (rover/drone)	17 / 17	29 / 12

Table 11: Comparison between Problem 5 original run and the Martian Hab topology

route consisting of three hops. As a result, the rover must perform several additional `move_via_tunnel` and `move_via_tunnel_to_beta` actions. Therefore, the larger number of rover actions is a consequence of the increased travel distance and does not represent different planning decisions.

The most informative result concerns `soil-sample-1`, the only artifact that could be assigned to either the rover or the drone, depending on the planner choice. In the generated plan, the rover collected it while travelling through Solar Farm, which was already part of its route for other required tasks. Instead of sending the drone specifically to collect this artifact, the planner exploited the opportunity to perform the pickup without additional travel cost. This demonstrates the cost aware reasoning that the topology change was trying to test. Rather than assigning the artifact to the agent that would be cheaper if considered in isolation, the planner considered the actions already required by the overall plan.

The result also shows that the drone’s relative speed advantage in the new topology did not cause it to take on a larger portion of the tasks. The tasks assigned to the drone, `bio-specimen-1` and `core-sample-1`, were already the ones most efficiently handled by flight in the original topology. Therefore, the topology change mainly affected travel costs and action counts, while the fundamental allocation of tasks between the rover and the drone remained largely unchanged.

8 Appendix - Plans

Below are reported some screens from the planners' runs. The plans and/or results are also saved in corresponding problems code folders.

8.1 Problem 1

```
[t=15.005269s, 34720 KB] Plan length: 34 step(s).
[t=15.005269s, 34720 KB] Plan cost: 34
[t=15.005269s, 34720 KB] Expanded 185969 state(s).
[t=15.005269s, 34720 KB] Reopened 2 state(s).
[t=15.005269s, 34720 KB] Evaluated 438377 state(s).
[t=15.005269s, 34720 KB] Evaluations: 438377
[t=15.005269s, 34720 KB] Generated 723689 state(s).
[t=15.005269s, 34720 KB] Dead ends: 0 state(s).
[t=15.005269s, 34720 KB] Expanded until last jump: 185955 state(s).
[t=15.005269s, 34720 KB] Reopened until last jump: 2 state(s).
[t=15.005269s, 34720 KB] Evaluated until last jump: 438309 state(s).
[t=15.005269s, 34720 KB] Generated until last jump: 723607 state(s).
[t=15.005269s, 34720 KB] Number of registered states: 438377
[t=15.005269s, 34720 KB] Int hash set load factor: 438377/1048576 = 0.418069
[t=15.005269s, 34720 KB] Int hash set resizes: 20
[t=15.005269s, 34720 KB] Search time: 14.969462s
[t=15.005269s, 34720 KB] Total time: 15.005269s
Solution found.
Peak memory: 34720 KB
Remove intermediate file output.sas
search exit code: 0

INFO    Planner time: 15.44s

move-via-tunnel curator1 tunnel stasis-lab (1)
putdown-artifact curator1 art-beta-1 stasis-lab (1)
[t=0.035335s, 12524 KB] Plan length: 35 step(s).
[t=0.035335s, 12524 KB] Plan cost: 35
[t=0.035335s, 12524 KB] Expanded 79 state(s).
[t=0.035335s, 12524 KB] Reopened 0 state(s).
[t=0.035335s, 12524 KB] Evaluated 80 state(s).
[t=0.035335s, 12524 KB] Evaluations: 160
[t=0.035335s, 12524 KB] Generated 374 state(s).
[t=0.035335s, 12524 KB] Dead ends: 0 state(s).
[t=0.035335s, 12524 KB] Number of registered states: 80
[t=0.035335s, 12524 KB] Int hash set load factor: 80/128 = 0.625000
[t=0.035335s, 12524 KB] Int hash set resizes: 7
[t=0.035335s, 12524 KB] Search time: 0.001613s
[t=0.035335s, 12524 KB] Total time: 0.035335s
Solution found.
Peak memory: 12524 KB
Remove intermediate file output.sas
search exit code: 0

INFO    Planner time: 0.48s
```

Figure 5: In order: Downward, Lama

8.2 Problem 2

```

[t=0.042511s, 12660 KB] Plan length: 31 step(s).
[t=0.042511s, 12660 KB] Plan cost: 31
[t=0.042511s, 12660 KB] Expanded 111 state(s).
[t=0.042511s, 12660 KB] Reopened 0 state(s).
[t=0.042511s, 12660 KB] Evaluated 112 state(s).
[t=0.042511s, 12660 KB] Evaluations: 224
[t=0.042511s, 12660 KB] Generated 1494 state(s).
[t=0.042511s, 12660 KB] Dead ends: 0 state(s).
[t=0.042511s, 12660 KB] Number of registered states: 112
[t=0.042511s, 12660 KB] Int hash set load factor: 112/128 = 0.875000
[t=0.042511s, 12660 KB] Int hash set resizes: 7
[t=0.042511s, 12660 KB] Search time: 0.003458s
[t=0.042511s, 12660 KB] Total time: 0.042511s
Solution found.
Peak memory: 12660 KB
Remove intermediate file output.sas
search exit code: 0

INFO      Planner time: 0.53s
root@942aac971015:/computer/Final_project/Autonomous-Preservation-Robots-in-Interpla
27: UNLOAD=FROM=POD CURATOR1 ART=ALPHA-2 PODZ CRTU=CHAMBER SLOT=CIA

time spent:    0.00 seconds instantiating 628 easy, 0 hard action templates
               0.00 seconds reachability analysis, yielding 96 facts and 379 actions
               0.00 seconds creating final representation with 93 relevant facts
               0.00 seconds building connectivity graph
               0.00 seconds searching, evaluating 58 states, to a max depth of 2
               0.00 seconds total time

root@942aac971015:/computer/Final_project/Autonomous-Preservation-Robots-in-Interplane

```

Figure 6: In order: Lama, FF

8.3 Problem 3

```
pandaPIparser is configured as follows
Colors in output: true
Mode: convert pandaPI plan
==>
1 activate-sealing curator1
31 move-via-tunnel curator1 entrance tunnel
140 move-via-tunnel-to-beta curator1 tunnel hall-beta
144 pickup-artifact curator1 art-beta-1 hall-beta slot-cla
153 move-via-tunnel-to-beta curator1 hall-beta tunnel
179 move-via-tunnel curator1 tunnel stasis-lab
226 putdown-artifact curator1 art-beta-1 stasis-lab slot-cla
291 move-via-tunnel curator1 stasis-lab tunnel
303 move-via-tunnel curator1 tunnel anti-vibration-pods
326 pickup-pod curator1 pod1 anti-vibration-pods slot-clb
339 move-via-tunnel curator1 anti-vibration-pods tunnel
349 move-via-tunnel-to-beta curator1 tunnel hall-beta
353 secure-in-pod curator1 art-beta-2 pod1 hall-beta slot-clb
372 move-via-tunnel-to-beta curator1 hall-beta tunnel
384 move-via-tunnel curator1 tunnel stasis-lab
401 unload-from-pod curator1 art-beta-2 pod1 stasis-lab slot-clb
424 move-via-tunnel curator1 stasis-lab tunnel
431 move-via-tunnel curator1 tunnel hall-alpha
484 pickup-artifact curator1 art-alpha-1 hall-alpha slot-cla
485 cool curator1 art-alpha-1 slot-cla
499 move-via-tunnel curator1 hall-alpha tunnel
566 move-via-tunnel curator1 tunnel cryo-chamber
577 putdown-artifact curator1 art-alpha-1 cryo-chamber slot-cla
599 move-via-tunnel curator1 cryo-chamber tunnel
610 move-via-tunnel curator1 tunnel anti-vibration-pods
630 pickup-pod curator1 pod2 anti-vibration-pods slot-cla
643 move-via-tunnel curator1 anti-vibration-pods tunnel
653 move-via-tunnel curator1 tunnel hall-alpha
676 secure-in-pod curator1 art-alpha-2 pod2 hall-alpha slot-cla
677 cool-in-pod curator1 art-alpha-2 pod2 slot-cla
691 move-via-tunnel curator1 hall-alpha tunnel
709 move-via-tunnel curator1 tunnel cryo-chamber
720 unload-from-pod curator1 art-alpha-2 pod2 cryo-chamber slot-cla
731 fly dronel entrance cryo-chamber
732 pickup-artifact dronel core-sample-1 cryo-chamber slot-d1a
742 fly dronel cryo-chamber stasis-lab
743 putdown-artifact dronel core-sample-1 stasis-lab slot-d1a
root 753
```

Figure 7: Panda

8.4 Problem 4

```

; Plan found with metric 40.007
; Theoretical reachable cost 40.009
; States evaluated so far: 109
; States pruned based on pre-heuristic cost lower bound: 0
; Time 0.17
0.000: (fly-to-beta drone1 entrance hall-beta) [2.000]
0.000: (activate-sealing curator1) [1.000]
1.001: (move-via-tunnel curator1 entrance tunnel) [3.000]
2.001: (pickup-artifact drone1 art-beta-1 hall-beta slot-d1a) [1.000]
3.001: (fly-from-beta drone1 hall-beta stasis-lab) [2.000]
4.002: (move-via-tunnel curator1 tunnel anti-vibration-pods) [3.000]
5.001: (putdown-artifact drone1 art-beta-1 stasis-lab slot-d1a) [1.000]
6.001: (fly drone1 stasis-lab hall-alpha) [2.000]
7.003: (pickup-pod curator1 pod1 anti-vibration-pods slot-c1a) [1.000]
8.001: (pickup-artifact drone1 art-alpha-1 hall-alpha slot-d1a) [1.000]
8.003: (move-via-tunnel curator1 anti-vibration-pods tunnel) [3.000]
9.001: (fly drone1 hall-alpha cryo-chamber) [2.000]
9.001: (cool drone1 art-alpha-1 slot-d1a) [3.000]
11.004: (move-via-tunnel-to-beta curator1 tunnel hall-beta) [3.000]
12.001: (putdown-artifact drone1 art-alpha-1 cryo-chamber slot-d1a) [1.000]
13.002: (pickup-artifact drone1 core-sample-1 cryo-chamber slot-d1a) [1.000]
14.002: (fly drone1 cryo-chamber stasis-lab) [2.000]
14.004: (secure-in-pod curator1 art-beta-2 pod1 hall-beta slot-c1a) [2.000]
16.002: (putdown-artifact drone1 core-sample-1 stasis-lab slot-d1a) [1.000]
16.004: (move-via-tunnel-to-beta curator1 hall-beta tunnel) [3.000]
19.005: (move-via-tunnel curator1 tunnel stasis-lab) [3.000]
22.005: (unload-from-pod curator1 art-beta-2 pod1 stasis-lab slot-c1a) [2.000]
24.005: (move-via-tunnel curator1 stasis-lab tunnel) [3.000]
27.006: (move-via-tunnel curator1 tunnel hall-alpha) [3.000]
30.006: (secure-in-pod curator1 art-alpha-2 pod1 hall-alpha slot-c1a) [2.000]
32.006: (move-via-tunnel curator1 hall-alpha tunnel) [3.000]
32.006: (cool-in-pod curator1 art-alpha-2 pod1 slot-c1a) [3.000]
35.007: (move-via-tunnel curator1 tunnel cryo-chamber) [3.000]
38.007: (unload-from-pod curator1 art-alpha-2 pod1 cryo-chamber slot-c1a) [2.000]

* All goal deadlines now no later than 40.007
Resorting to best-first search
Running WA* with W = 5.000, not restarting with goal states
b (23.000 | 2.000)b (23.000 | 1.000)b (22.000 | 2.000)b (21.000 | 4.001)b (20.000 | 5.000)b (19.000 | 6.000)b
(18.000 | 7.002)002)b (16.000 | 8.000)b (15.000 | 11.001)b (14.000 | 12.001)b (13.000 | 12.001)b (12.000 | 13.004)b
(11.000 | 15.004)b (10.000.000 | 25.005)b (8.000 | 30.006)b (7.000 | 33.006)b (7.000 | 22.005)b (6.000 | 24.005)b
(5.000 | 27.005)b (4.000 | 32.006)b (3b (3.000 | 32.006)Killed

```

Figure 8: Non-seismic scenario. In order: OPTIC

```

42.25000000: (move-via-tunnel curator1 cryo-chamber tunnel) [3.00000000]
45.26000000: (move-via-tunnel-to-beta curator1 tunnel hall-beta) [3.00000000]
48.27000000: (secure-in-pod curator1 art-beta-2 pod1 hall-beta slot-clb) [2.00000000]
50.28000000: (move-via-tunnel-to-beta curator1 hall-beta tunnel) [3.00000000]
53.29000000: (move-via-tunnel curator1 tunnel stasis-lab) [3.00000000]
56.30000000: (unload-from-pod curator1 art-beta-2 pod1 stasis-lab slot-clb) [2.00000000]
Rescheduled Plan:
0.00100000: (fly dronel entrance hall-alpha) [2.00000000]
2.01100000: (pickup-artifact dronel art-alpha-1 hall-alpha slot-d1a) [1.00000000]
3.02100000: (fly dronel hall-alpha cryo-chamber) [2.00000000]
5.03100000: (putdown-artifact dronel art-alpha-1 cryo-chamber slot-d1a) [1.00000000]
6.04100000: (pickup-artifact dronel core-sample-1 cryo-chamber slot-d1a) [1.00000000]
7.05100000: (fly dronel cryo-chamber stasis-lab) [2.00000000]
9.06100000: (putdown-artifact dronel core-sample-1 stasis-lab slot-d1a) [1.00000000]
10.07100000: (fly-to-beta dronel stasis-lab hall-beta) [2.00000000]
12.08100000: (pickup-artifact dronel art-beta-1 hall-beta slot-d1a) [1.00000000]
13.09100000: (fly-from-beta dronel hall-beta stasis-lab) [2.00000000]
15.10100000: (putdown-artifact dronel art-beta-1 stasis-lab slot-d1a) [1.00000000]
0.00100000: (activate-sealing curator1) [1.00000000]
1.01100000: (move-via-tunnel curator1 entrance tunnel) [3.00000000]
16.11100000: (fly dronel stasis-lab cryo-chamber) [2.00000000]
4.02100000: (move-via-tunnel curator1 tunnel anti-vibration-pods) [3.00000000]
7.03100000: (pickup-pod curator1 pod1 anti-vibration-pods slot-clb) [1.00000000]
8.04100000: (move-via-tunnel curator1 anti-vibration-pods tunnel) [3.00000000]
11.05100000: (move-via-tunnel curator1 tunnel hall-alpha) [3.00000000]
14.06100000: (secure-in-pod curator1 art-alpha-2 pod1 hall-alpha slot-clb) [2.00000000]
16.07100000: (cool-in-pod curator1 art-alpha-2 pod1 slot-clb) [3.00000000]
16.08100000: (move-via-tunnel curator1 hall-alpha tunnel) [3.00000000]
18.12100000: (pickup-artifact dronel art-alpha-1 cryo-chamber slot-d1a) [1.00000000]
19.13100000: (cool dronel art-alpha-1 slot-d1a) [3.00000000]
22.14100000: (putdown-artifact dronel art-alpha-1 cryo-chamber slot-d1a) [1.00000000]
19.09100000: (move-via-tunnel curator1 tunnel cryo-chamber) [3.00000000]
22.10100000: (unload-from-pod curator1 art-alpha-2 pod1 cryo-chamber slot-clb) [2.00000000]
24.11100000: (move-via-tunnel curator1 cryo-chamber tunnel) [3.00000000]
27.12100000: (move-via-tunnel-to-beta curator1 tunnel hall-beta) [3.00000000]
30.13100000: (secure-in-pod curator1 art-beta-2 pod1 hall-beta slot-clb) [2.00000000]
32.14100000: (move-via-tunnel-to-beta curator1 hall-beta tunnel) [3.00000000]
35.15100000: (move-via-tunnel curator1 tunnel stasis-lab) [3.00000000]
38.16100000: (unload-from-pod curator1 art-beta-2 pod1 stasis-lab slot-clb) [2.00000000]
Solution with original makespan 58.3 found (ignoring no-moving-targets-rule).
Plan length: 32 step(s).
Makespan : 40.161
Rescheduled Makespan : 40.161
Search time: 0.09 seconds - Walltime: 0.0910459 seconds
Total time: 0.09 seconds - Walltime: 0.0942359 seconds

```

```

Number of literals: 93
Constructing lookup tables: [10%] [20%] [30%] [40%] [50%] [60%] [70%] [80%] [90%] [100%]
Post filtering unreachable actions: [10%] [20%] [30%] [40%] [50%] [60%] [70%] [80%] [90%] [100%]
No analytic limits found, not considering limit effects of goal-only operators
All the ground actions in this problem are compression-safe
Initial heuristic = 24.000
b (23.000 | 2.000)b (22.000 | 2.000)b (21.000 | 4.001)b (17.000 | 5.001)b (16.000 | 6.001)b (15.000 | 11.001)b (14.000 | 12.001)b
; States evaluated: 122
; Cost: 54.009
; Time 0.12
0.000: (fly-to-beta dronel entrance hall-beta) [2.000]
0.000: (activate-sealing curator1) [1.000]
1.001: (move-via-tunnel curator1 entrance tunnel) [3.000]
2.001: (pickup-artifact dronel art-beta-1 hall-beta slot-d1a) [1.000]
3.001: (fly-from-beta dronel hall-beta stasis-lab) [2.000]
4.002: (move-via-tunnel curator1 tunnel hall-alpha) [3.000]
5.001: (putdown-artifact dronel art-beta-1 stasis-lab slot-d1a) [1.000]
6.001: (fly dronel stasis-lab cryo-chamber) [2.000]
7.003: (pickup-artifact curator1 art-alpha-1 hall-alpha slot-cla) [1.000]
8.001: (pickup-artifact dronel core-sample-1 cryo-chamber slot-d1a) [1.000]
8.003: (cool curator1 art-alpha-1 slot-cla) [3.000]
8.003: (move-via-tunnel curator1 hall-alpha tunnel) [3.000]
9.001: (fly dronel cryo-chamber stasis-lab) [2.000]
11.001: (putdown-artifact dronel core-sample-1 stasis-lab slot-d1a) [1.000]
11.004: (move-via-tunnel curator1 tunnel cryo-chamber) [3.000]
14.004: (putdown-artifact curator1 art-alpha-1 cryo-chamber slot-cla) [1.000]
15.004: (move-via-tunnel curator1 cryo-chamber tunnel) [3.000]
18.005: (move-via-tunnel curator1 tunnel anti-vibration-pods) [3.000]
21.005: (pickup-pod curator1 pod1 anti-vibration-pods slot-cla) [1.000]
22.005: (move-via-tunnel curator1 anti-vibration-pods tunnel) [3.000]
25.006: (move-via-tunnel-to-beta curator1 tunnel hall-beta) [3.000]
28.006: (secure-in-pod curator1 art-beta-2 pod1 hall-beta slot-cla) [2.000]
30.006: (move-via-tunnel-to-beta curator1 hall-beta tunnel) [3.000]
33.007: (move-via-tunnel curator1 tunnel stasis-lab) [3.000]
36.007: (unload-from-pod curator1 art-beta-2 pod1 stasis-lab slot-cla) [2.000]
38.007: (move-via-tunnel curator1 stasis-lab tunnel) [3.000]
41.008: (move-via-tunnel curator1 tunnel hall-alpha) [3.000]
44.008: (secure-in-pod curator1 art-alpha-2 pod1 hall-alpha slot-cla) [2.000]
46.008: (move-via-tunnel curator1 hall-alpha tunnel) [3.000]
46.008: (cool-in-pod curator1 art-alpha-2 pod1 slot-cla) [3.000]
49.009: (move-via-tunnel curator1 tunnel cryo-chamber) [3.000]
52.009: (unload-from-pod curator1 art-alpha-2 pod1 cryo-chamber slot-cla) [2.000]

```

Figure 9: Non-seismic scenario. In order: TFD and POPF

```

; Plan found with metric 45.001
; Theoretical reachable cost 45.003
; States evaluated so far: 2144
; States pruned based on pre-heuristic cost lower bound: 0
; Time 1.72
0.000: (fly dronel entrance cryo-chamber) [2.000]
0.000: (activate-sealing curator1) [1.000]
1.001: (move-via-tunnel curator1 entrance tunnel) [3.000]
2.001: (pickup-artifact dronel core-sample-1 cryo-chamber slot-d1a) [1.000]
3.001: (fly dronel cryo-chamber stasis-lab) [2.000]
4.002: (move-via-tunnel curator1 tunnel anti-vibration-pods) [3.000]
5.001: (putdown-artifact dronel core-sample-1 stasis-lab slot-d1a) [1.000]
6.001: (fly dronel stasis-lab hall-alpha) [2.000]
7.003: (pickup-pod curator1 pod1 anti-vibration-pods slot-c1a) [1.000]
8.001: (pickup-artifact dronel art-alpha-1 hall-alpha slot-d1a) [1.000]
8.003: (move-via-tunnel curator1 anti-vibration-pods tunnel) [3.000]
9.001: (fly dronel hall-alpha cryo-chamber) [2.000]
11.004: (move-via-tunnel curator1 tunnel hall-alpha) [3.000]
11.998: (cool dronel art-alpha-1 slot-d1a) [3.000]
14.004: (secure-in-pod curator1 art-alpha-2 pod1 hall-alpha slot-c1a) [2.000]
14.998: (putdown-artifact dronel art-alpha-1 cryo-chamber slot-d1a) [1.000]
15.998: (fly dronel cryo-chamber anti-vibration-pods) [2.000]
16.004: (move-via-tunnel curator1 hall-alpha tunnel) [3.000]
17.999: (fly dronel anti-vibration-pods stasis-lab) [2.000]
19.005: (cool-in-pod curator1 art-alpha-2 pod1 slot-c1a) [3.000]
19.005: (move-via-tunnel curator1 tunnel cryo-chamber) [3.000]
20.000: (fly-to-beta dronel stasis-lab hall-beta) [2.000]
22.000: (pickup-artifact dronel art-beta-1 hall-beta slot-d1a) [1.000]
22.005: (unload-from-pod curator1 art-alpha-2 pod1 cryo-chamber slot-c1a) [2.000]
23.000: (fly-from-beta dronel hall-beta stasis-lab) [2.000]
25.000: (putdown-artifact dronel art-beta-1 stasis-lab slot-d1a) [1.000]
28.999: (move-via-tunnel curator1 cryo-chamber tunnel) [3.000]
32.000: (move-via-tunnel-to-beta curator1 tunnel hall-beta) [3.000]
35.000: (secure-in-pod curator1 art-beta-2 pod1 hall-beta slot-c1a) [2.000]
37.000: (move-via-tunnel-to-beta curator1 hall-beta tunnel) [3.000]
40.001: (move-via-tunnel curator1 tunnel stasis-lab) [3.000]
43.001: (unload-from-pod curator1 art-beta-2 pod1 stasis-lab slot-c1a) [2.000]

* All goal deadlines now no later than 45.001

```

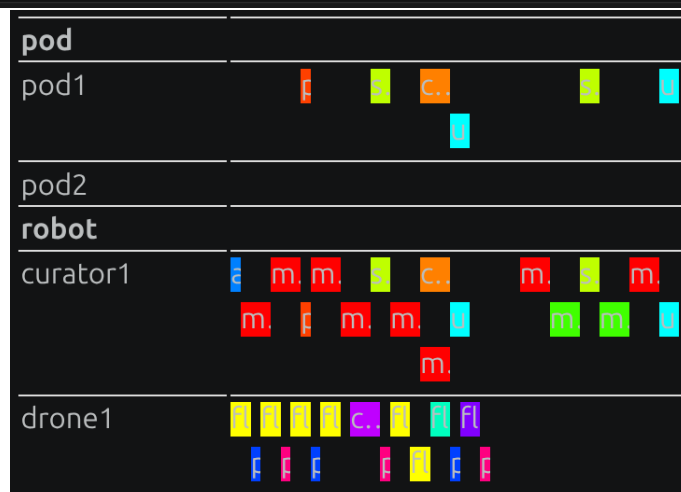


Figure 10: Seismic scenario. In order: OPTIC; scheme of actions generated from the plan with [1]

8.5 Problem 5

```

> get plan
plan:
0:      (activate_sealing curator1)      [1]
0:      (fly dronel entrance anti-vibration-pods) [2]
1.001:  (move_via_tunnel curator1 entrance tunnel) [3]
2.001:  (fly dronel anti-vibration-pods hall-alpha) [2]
4.002:  (fly dronel hall-alpha stasis-lab) [2]
4.002:  (move_via_tunnel curator1 tunnel anti-vibration-pods) [3]
6.003:  (fly dronel stasis-lab cryo-chamber) [2]
7.003:  (pickup_pod curator1 pod1 anti-vibration-pods slot-cla) [1]
8.003:  (pickup_artifact dronel core-sample-1 cryo-chamber slot-d1a) [1]
8.003:  (move_via_tunnel curator1 anti-vibration-pods tunnel) [3]
9.003:  (fly dronel cryo-chamber stasis-lab) [2]
11.003: (putdown_artifact dronel core-sample-1 stasis-lab slot-d1a) [1]
11.004: (move_via_tunnel curator1 tunnel hall-alpha) [3]
12.003: (fly dronel stasis-lab hall-alpha) [2]
14.003: (pickup_artifact dronel art-alpha-1 hall-alpha slot-d1a) [1]
14.004: (secure_in_pod curator1 art-alpha-2 pod1 hall-alpha slot-cla) [2]
15.003: (cool dronel art-alpha-1 slot-d1a hall-alpha) [3]
16.004: (cool_in_pod curator1 art-alpha-2 pod1 slot-cla hall-alpha) [3]
18.003: (fly dronel hall-alpha cryo-chamber) [2]
19.004: (move_via_tunnel curator1 hall-alpha tunnel) [3]
20.003: (putdown_artifact dronel art-alpha-1 cryo-chamber slot-d1a) [1]
21.003: (fly dronel cryo-chamber stasis-lab) [2]
22.005: (move_via_tunnel curator1 tunnel cryo-chamber) [3]
23.004: (fly_to_beta dronel stasis-lab hall-beta) [2]
25.004: (pickup_artifact dronel art-beta-1 hall-beta slot-d1a) [1]
25.005: (unload_from_pod curator1 art-alpha-2 pod1 cryo-chamber slot-cla) [2]
26.004: (fly_from_beta dronel hall-beta stasis-lab) [2]
27.005: (move_via_tunnel curator1 cryo-chamber tunnel) [3]
28.004: (putdown_artifact dronel art-beta-1 stasis-lab slot-d1a) [1]
30.006: (move_via_tunnel_to_beta curator1 tunnel hall-beta) [3]
33.006: (secure_in_pod curator1 art-beta-2 pod1 hall-beta slot-cla) [2]
35.006: (move_via_tunnel_to_beta curator1 hall-beta tunnel) [3]
38.007: (move_via_tunnel curator1 tunnel stasis-lab) [3]
41.007: (unload_from_pod curator1 art-beta-2 pod1 stasis-lab slot-cla) [2]

> run
[(move_via_tunnel curator1 anti-vibration-pods tunnel) 13.3333%][(pickup_artifact dronel core-sample-1 cryo-chamber slot-d1a)
[(move_via_tunnel curator1 anti-vibration-pods tunnel) 26.6667%][(pickup_artifact dronel core-sample-1 cryo-chamber slot-d1a)
[(move_via_tunnel curator1 anti-vibration-pods tunnel) 33.3333%][(pickup_artifact dronel core-sample-1 cryo-chamber slot-d1a)
[(move_via_tunnel curator1 anti-vibration-pods tunnel) 46.6667%][(pickup_artifact dronel core-sample-1 cryo-chamber slot-d1a)
[(pickup_artifact dronel art-alpha-1 hall-alpha slot-d1a) 0%][(secure_in_pod curator1 art-alpha-2 pod1 hall-alpha slot-cla) 5
[(pickup_artifact dronel art-alpha-1 hall-alpha slot-d1a) 20%][(secure_in_pod curator1 art-alpha-2 pod1 hall-alpha slot-cla)
[(pickup_artifact dronel art-alpha-1 hall-alpha slot-d1a) 40%][(secure_in_pod curator1 art-alpha-2 pod1 hall-alpha slot-cla)
[(pickup_artifact dronel art-alpha-1 hall-alpha slot-d1a) 60%][(secure_in_pod curator1 art-alpha-2 pod1 hall-alpha slot-cla)
[(pickup_artifact dronel art-alpha-1 hall-alpha slot-d1a) 80%][(secure_in_pod curator1 art-alpha-2 pod1 hall-alpha slot-cla)
[(pickup_artifact dronel art-alpha-1 hall-alpha slot-d1a) 100%][(secure_in_pod curator1 art-alpha-2 pod1 hall-alpha slot-cla)
[(cool dronel art-alpha-1 slot-d1a hall-alpha) 13.3333%][(cool_in_pod curator1 art-alpha-2 pod1 slot-cla hall-alpha) 6.66667%
[(cool dronel art-alpha-1 slot-d1a hall-alpha) 33.3333%][(cool_in_pod curator1 art-alpha-2 pod1 slot-cla hall-alpha) 26.6667%
[(cool dronel art-alpha-1 slot-d1a hall-alpha) 53.3333%][(cool_in_pod curator1 art-alpha-2 pod1 slot-cla hall-alpha) 46.6667%
[(cool dronel art-alpha-1 slot-d1a hall-alpha) 73.3333%][(cool_in_pod curator1 art-alpha-2 pod1 slot-cla hall-alpha) 66.6667%
[(cool dronel art-alpha-1 slot-d1a hall-alpha) 93.3333%][(cool_in_pod curator1 art-alpha-2 pod1 slot-cla hall-alpha) 86.6667%
[(pickup_artifact dronel art-beta-1 hall-beta slot-d1a) 0%][(unload_from_pod curator1 art-alpha-2 pod1 cryo-chamber slot-cla)
[(pickup_artifact dronel art-beta-1 hall-beta slot-d1a) 20%][(unload_from_pod curator1 art-alpha-2 pod1 cryo-chamber slot-cla)
[(pickup_artifact dronel art-beta-1 hall-beta slot-d1a) 40%][(unload_from_pod curator1 art-alpha-2 pod1 cryo-chamber slot-cla)
[(pickup_artifact dronel art-beta-1 hall-beta slot-d1a) 60%][(unload_from_pod curator1 art-alpha-2 pod1 cryo-chamber slot-cla)
[INFO] [1788767634.235635818] [executor_client]: Plan Succeeded

Successful finished
> 

```

Figure 11: Plan and run on Terminal 2

```
As soon as any node with that name is destructed it will unregister the publisher, preventing any further logs for that name
from being published on the rosout topic.
activate_sealing ... [100%]
fly ... [100%]
move_via_tunnel ... [100%]
fly ... [100%]
fly ... [100%]
move_via_tunnel ... [100%]
pickup_pod ... [100%]
fly ... [100%]
pickup_artifact ... [100%]
[plansys2_node-1] [WARN] [1788767592.207522045] [LifecyclePublisher]: Trying to publish message on the topic '/actions_hub',
but the publisher is not activated
move_via_tunnel ... [100%]
fly ... [100%]
putdown_artifact ... [100%]
move_via_tunnel ... [100%]
fly ... [100%]
pickup_artifact ... [100%]
secure_in_pod ... [100%]
cool ... [100%]
cool_in_pod ... [100%]
fly ... [100%]
move_via_tunnel ... [100%]
putdown_artifact ... [100%]
fly ... [100%]
move_via_tunnel ... [100%]
fly_to_beta ... [100%]
unload_from_pod ... [100%]
pickup_artifact ... [100%]
fly_from_beta ... [100%]
move_via_tunnel ... [100%]
putdown_artifact ... [100%]
move_via_tunnel_to_beta ... [100%]
secure_in_pod ... [100%]
move_via_tunnel_to_beta ... [100%]
move_via_tunnel ... [100%]
unload_from_pod ... [100%]
[plansys2_node-1] [INFO] [1788767632.101466699] [executor]: Plan Succeeded
□
```

Figure 12: Execution on Terminal 1

8.6 Problem 5 - Bonus

```

plan:
0:      (activate_sealing rover1)          [1]
0:      (fly recon_drone1 airlock hydroponics-bay) [2]
1.001:  (move_via_tunnel rover1 airlock solar-farm) [3]
2.001:  (pickup_artifact recon_drone1 bio-specimen-1 hydroponics-bay slot-d1a) [1]
3.002:  (cool recon_drone1 bio-specimen-1 slot-d1a hydroponics-bay) [3]
4.002:  (pickup_artifact rover1 soil-sample-1 solar-farm slot-r1a) [1]
5.002:  (move_via_tunnel rover1 solar-farm airlock) [3]
6.002:  (putdown_artifact recon_drone1 bio-specimen-1 hydroponics-bay slot-d1a) [1]
7.002:  (fly recon_drone1 hydroponics-bay med-bay) [2]
8.003:  (move_via_tunnel rover1 airlock hab-corridor) [3]
9.002:  (pickup_artifact recon_drone1 core-sample-1 med-bay slot-d1a) [1]
10.002: (fly recon_drone1 med-bay sample-locker) [2]
11.004: (move_via_tunnel rover1 hab-corridor sample-locker) [3]
12.002: (putdown_artifact recon_drone1 core-sample-1 sample-locker slot-d1a) [1]
13.002: (fly recon_drone1 sample-locker hydroponics-bay) [2]
14.004: (putdown_artifact rover1 soil-sample-1 sample-locker slot-r1a) [1]
15.002: (pickup_artifact recon_drone1 bio-specimen-1 hydroponics-bay slot-d1a) [1]
15.004: (move_via_tunnel rover1 sample-locker hab-corridor) [3]
16.002: (fly recon_drone1 hydroponics-bay med-bay) [2]
18.002: (putdown_artifact recon_drone1 bio-specimen-1 med-bay slot-d1a) [1]
18.005: (move_via_tunnel rover1 hab-corridor workshop) [3]
21.005: (pickup_pod rover1 cratel workshop slot-r1a) [1]
22.005: (move_via_tunnel rover1 workshop hab-corridor) [3]
25.006: (move_via_tunnel rover1 hab-corridor hydroponics-bay) [3]
28.006: (secure_in_pod rover1 bio-specimen-2 cratel hydroponics-bay slot-r1a) [2]
30.006: (cool_in_pod rover1 bio-specimen-2 cratel slot-r1a hydroponics-bay) [3]
33.006: (move_via_tunnel rover1 hydroponics-bay hab-corridor) [3]
36.007: (move_via_tunnel rover1 hab-corridor med-bay) [3]
39.007: (unload_from_pod rover1 bio-specimen-2 cratel med-bay slot-r1a) [2]
41.007: (move_via_tunnel rover1 med-bay hab-corridor) [3]
44.008: (move_via_tunnel rover1 hab-corridor airlock) [3]
47.009: (move_via_tunnel rover1 airlock waypoint-1) [3]
50.01:  (move_via_tunnel rover1 waypoint-1 waypoint-2) [3]
53.011: (move_via_tunnel_to_beta rover1 waypoint-2 pathfinder-site) [3]
56.011: (secure_in_pod rover1 soil-sample-2 cratel pathfinder-site slot-r1a) [2]
58.011: (move_via_tunnel_to_beta rover1 pathfinder-site waypoint-2) [3]
61.012: (move_via_tunnel rover1 waypoint-2 waypoint-1) [3]
64.013: (move_via_tunnel rover1 waypoint-1 airlock) [3]
67.014: (move_via_tunnel rover1 airlock hab-corridor) [3]
70.015: (move_via_tunnel rover1 hab-corridor sample-locker) [3]
73.015: (unload_from_pod rover1 soil-sample-2 cratel sample-locker slot-r1a) [2]

> run
[(move_via_tunnel rover1 airlock solar-farm) 46.6667%][(pickup_artifact recon_drone1 bio-specimen-1 hydroponics-bay slot-d1a)
[(move_via_tunnel rover1 airlock solar-farm) 53.3333%][(pickup_artifact recon_drone1 bio-specimen-1 hydroponics-bay slot-d1a)
[(move_via_tunnel rover1 airlock solar-farm) 66.6667%][(pickup_artifact recon_drone1 bio-specimen-1 hydroponics-bay slot-d1a)
[(move_via_tunnel rover1 airlock solar-farm) 73.3333%][(pickup_artifact recon_drone1 bio-specimen-1 hydroponics-bay slot-d1a)
[(cool recon_drone1 bio-specimen-1 slot-d1a hydroponics-bay) 33.3333%][(pickup_artifact rover1 soil-sample-1 solar-farm slot-
[(cool recon_drone1 bio-specimen-1 slot-d1a hydroponics-bay) 40%][(pickup_artifact rover1 soil-sample-1 solar-farm slot-r1a)
[(cool recon_drone1 bio-specimen-1 slot-d1a hydroponics-bay) 46.6667%][(pickup_artifact rover1 soil-sample-1 solar-farm slot-
[(cool recon_drone1 bio-specimen-1 slot-d1a hydroponics-bay) 53.3333%][(pickup_artifact rover1 soil-sample-1 solar-farm slot-
[(cool recon_drone1 bio-specimen-1 slot-d1a hydroponics-bay) 60%][(pickup_artifact rover1 soil-sample-1 solar-farm slot-r1a)
[(cool recon_drone1 bio-specimen-1 slot-d1a hydroponics-bay) 66.6667%][(pickup_artifact rover1 soil-sample-1 solar-farm slot-
[(move_via_tunnel rover1 solar-farm airlock) 46.6667%][(putdown_artifact recon_drone1 bio-specimen-1 hydroponics-bay slot-d1a)
[(move_via_tunnel rover1 solar-farm airlock) 53.3333%][(putdown_artifact recon_drone1 bio-specimen-1 hydroponics-bay slot-d1a)
[(move_via_tunnel rover1 solar-farm airlock) 60%][(putdown_artifact recon_drone1 bio-specimen-1 hydroponics-bay slot-d1a) 60%
[(move_via_tunnel rover1 solar-farm airlock) 66.6667%][(putdown_artifact recon_drone1 bio-specimen-1 hydroponics-bay slot-d1a)
[(move_via_tunnel rover1 solar-farm airlock) 73.3333%][(putdown_artifact recon_drone1 bio-specimen-1 hydroponics-bay slot-d1a)
[(move_via_tunnel rover1 hab-corridor sample-locker) 60%][(putdown_artifact recon_drone1 core-sample-1 sample-locker slot-d1a)
[(move_via_tunnel rover1 hab-corridor sample-locker) 66.6667%][(putdown_artifact recon_drone1 core-sample-1 sample-locker slo
[(move_via_tunnel rover1 hab-corridor sample-locker) 73.3333%][(putdown_artifact recon_drone1 core-sample-1 sample-locker slo
[(move_via_tunnel rover1 hab-corridor sample-locker) 80%][(putdown_artifact recon_drone1 core-sample-1 sample-locker slot-d1a)
[(move_via_tunnel rover1 hab-corridor sample-locker) 86.6667%][(putdown_artifact recon_drone1 core-sample-1 sample-locker slo
[(move_via_tunnel rover1 hab-corridor sample-locker) 93.3333%][(putdown_artifact recon_drone1 core-sample-1 sample-locker slo
[(move_via_tunnel rover1 sample-locker hab-corridor) 26.6667%][(pickup_artifact recon_drone1 bio-specimen-1 hydroponics-bay s
[(move_via_tunnel rover1 sample-locker hab-corridor) 33.3333%][(pickup_artifact recon_drone1 bio-specimen-1 hydroponics-bay s
[(move_via_tunnel rover1 sample-locker hab-corridor) 40%][(pickup_artifact recon_drone1 bio-specimen-1 hydroponics-bay slot-d
[(move_via_tunnel rover1 sample-locker hab-corridor) 46.6667%][(pickup_artifact recon_drone1 bio-specimen-1 hydroponics-bay s
[(move_via_tunnel rover1 sample-locker hab-corridor) 53.3333%][(pickup_artifact recon_drone1 bio-specimen-1 hydroponics-bay s
[(move_via_tunnel rover1 sample-locker hab-corridor) 60%][(pickup_artifact recon_drone1 bio-specimen-1 hydroponics-bay slot-d
[(move_via_tunnel rover1 hab-corridor workshop) 46.6667%][(putdown_artifact recon_drone1 bio-specimen-1 med-bay slot-d1a) 20%
[(move_via_tunnel rover1 hab-corridor workshop) 53.3333%][(putdown_artifact recon_drone1 bio-specimen-1 med-bay slot-d1a) 40%
[(move_via_tunnel rover1 hab-corridor workshop) 66.6667%][(putdown_artifact recon_drone1 bio-specimen-1 med-bay slot-d1a) 80%
[(move_via_tunnel rover1 hab-corridor workshop) 73.3333%][(putdown_artifact recon_drone1 bio-specimen-1 med-bay slot-d1a) 100
[INFO] [1788766660.937916479] [executor_client]: Plan Succeeded

Successful finished
>

```

Figure 13: Plan and run on Terminal 2

```
move_via_tunnel ... [100%]
[plansys2_node-1] [WARN] [1788766579.499809858] [LifecyclePublisher]: Trying to publish message on the topic '/actions_hub',
but the publisher is not activated
fly ... [100%]
pickup_artifact ... [100%]
move_via_tunnel ... [100%]
[plansys2_node-1] [WARN] [1788766583.001107628] [LifecyclePublisher]: Trying to publish message on the topic '/actions_hub',
but the publisher is not activated
fly ... [100%]
putdown_artifact ... [100%]
move_via_tunnel ... [100%]
putdown_artifact ... [100%]
fly ... [100%]
pickup_artifact ... [100%]
move_via_tunnel ... [100%]
fly ... [100%]
putdown_artifact ... [100%]
move_via_tunnel ... [100%]
pickup_pod ... [100%]
move_via_tunnel ... [100%]
move_via_tunnel ... [100%]
secure_in_pod ... [100%]
cool_in_pod ... [100%]
move_via_tunnel ... [100%]
move_via_tunnel ... [100%]
unload_from_pod ... [100%]
move_via_tunnel ... [100%]
move_via_tunnel ... [100%]
move_via_tunnel ... [100%]
move_via_tunnel ... [100%]
move_via_tunnel_to_beta ... [100%]
secure_in_pod ... [100%]
move_via_tunnel_to_beta ... [100%]
move_via_tunnel ... [100%]
move_via_tunnel ... [100%]
move_via_tunnel ... [100%]
move_via_tunnel ... [100%]
unload_from_pod ... [100%]
[plansys2_node-1] [INFO] [1788766658.695582301] [executor]: Plan Succeeded
```

Figure 14: Execution on Terminal 1

8.7 Additional Problems

8.7.1 Scaled Artifact Count

```

fly-to-beta drone1 stasis-lab hall-beta (1)
pickup-artifact drone1 art-beta-1 hall-beta slot-d1a (1)
fly-from-beta drone1 hall-beta stasis-lab (1)
putdown-artifact drone1 art-beta-1 stasis-lab slot-d1a (1)
fly-to-beta drone1 stasis-lab hall-beta (1)
pickup-artifact drone1 art-beta-4 hall-beta slot-d1a (1)
fly-from-beta drone1 hall-beta stasis-lab (1)
putdown-artifact drone1 art-beta-4 stasis-lab slot-d1a (1)
[t=0.054091s, 12792 KB] Plan length: 51 step(s).
[t=0.054091s, 12792 KB] Plan cost: 51
[t=0.054091s, 12792 KB] Expanded 192 state(s).
[t=0.054091s, 12792 KB] Reopened 0 state(s).
[t=0.054091s, 12792 KB] Evaluated 193 state(s).
[t=0.054091s, 12792 KB] Evaluations: 386
[t=0.054091s, 12792 KB] Generated 2849 state(s).
[t=0.054091s, 12792 KB] Dead ends: 0 state(s).
[t=0.054091s, 12792 KB] Number of registered states: 193
[t=0.054091s, 12792 KB] Int hash set load factor: 193/256 = 0.753906
[t=0.054091s, 12792 KB] Int hash set resizes: 8
[t=0.054091s, 12792 KB] Search time: 0.007699s
[t=0.054091s, 12792 KB] Total time: 0.054091s
Solution found.
Peak memory: 12792 KB
Remove intermediate file output.sas
search exit code: 0

INFO      Planner time: 0.57s

27: PUTDOWN-ARTIFACT CURATOR1 ART-ALPHA-1 CRYO-CHAMBER SLOT-C1B
28: FLY-TO-BETA DRONE1 CRYO-CHAMBER HALL-BETA
29: UNLOAD-FROM-POD CURATOR1 ART-ALPHA-2 POD2 CRYO-CHAMBER SLOT-C1A
30: UNLOAD-FROM-POD CURATOR1 ART-ALPHA-3 POD2 CRYO-CHAMBER SLOT-C1A
31: PICKUP-ARTIFACT CURATOR1 CORE-SAMPLE-2 CRYO-CHAMBER SLOT-C1B
32: MOVE-VIA-TUNNEL CURATOR1 CRYO-CHAMBER TUNNEL
33: FLY-FROM-BETA DRONE1 HALL-BETA STASIS-LAB
34: MOVE-VIA-TUNNEL CURATOR1 TUNNEL STASIS-LAB
35: UNLOAD-FROM-POD CURATOR1 ART-BETA-2 POD2 STASIS-LAB SLOT-C1A
36: UNLOAD-FROM-POD CURATOR1 ART-BETA-3 POD2 STASIS-LAB SLOT-C1A
37: PUTDOWN-ARTIFACT DRONE1 CORE-SAMPLE-1 STASIS-LAB SLOT-D1A
38: PUTDOWN-ARTIFACT CURATOR1 CORE-SAMPLE-2 STASIS-LAB SLOT-C1B
39: FLY-TO-BETA DRONE1 STASIS-LAB HALL-BETA
40: PICKUP-ARTIFACT DRONE1 ART-BETA-1 HALL-BETA SLOT-D1A
41: FLY-FROM-BETA DRONE1 HALL-BETA STASIS-LAB
42: PUTDOWN-ARTIFACT DRONE1 ART-BETA-1 STASIS-LAB SLOT-D1A
43: FLY-TO-BETA DRONE1 STASIS-LAB HALL-BETA
44: PICKUP-ARTIFACT DRONE1 ART-BETA-4 HALL-BETA SLOT-D1A
45: FLY-FROM-BETA DRONE1 HALL-BETA STASIS-LAB
46: PUTDOWN-ARTIFACT DRONE1 ART-BETA-4 STASIS-LAB SLOT-D1A

time spent:    0.00 seconds instantiating 1108 easy, 0 hard action templates
               0.00 seconds reachability analysis, yielding 153 facts and 624 actions
               0.00 seconds creating final representation with 147 relevant facts
               0.00 seconds building connectivity graph
               0.00 seconds searching, evaluating 122 states, to a max depth of 3

```

Figure 15: In order: Lama, FF

```

[t=77.012840s, 40992 KB] g=25, 456631 evaluated, 64963 expanded
[t=341.916451s, 121816 KB] f = 38, 2039896 evaluated, 302950 expanded, 965 reopened
[t=341.956563s, 121816 KB] New best heuristic value for lmcut: 11
[t=341.956696s, 121816 KB] g=27, 2040281 evaluated, 302991 expanded, 965 reopened
[t=343.904365s, 121816 KB] New best heuristic value for lmcut: 10
[t=343.904487s, 121816 KB] g=27, 2057239 evaluated, 305376 expanded, 965 reopened
[t=1429.368750s, 452608 KB] f = 39, 8742221 evaluated, 1320387 expanded, 2178 reopened
[t=1429.383369s, 452608 KB] New best heuristic value for lmcut: 9
[t=1429.383426s, 452608 KB] g=30, 8742396 evaluated, 1320404 expanded, 2178 reopened
[t=1471.012817s, 465484 KB] New best heuristic value for lmcut: 8
[t=1471.012938s, 465484 KB] g=31, 9107495 evaluated, 1368203 expanded, 2202 reopened

^CPeak memory: 1750148 KB
caught signal 2 -- exiting
Traceback (most recent call last):
  File "/workspace/downward/fast-downward.py", line 5, in <module>
    main()

root@faeb39914efb:/computer/Final_project/Autonomous-Preservation-Robots-in-Interplanetary-Museum-Vault-scen
Additional_Problems# planutils run val -- Validate ../Problem_2/domain.pddl problem_scaleup.pddl problem_sca
pddl.plan
Checking plan: problem_scaleup.pddl.plan
Plan executed successfully - checking goal
Plan valid
Final value: 47

Successful plans:
Value: 47
  problem_scaleup.pddl.plan 47

root@faeb39914efb:/computer/Final_project/Autonomous-Preservation-Robots-in-Interplanetary-Museum-Vault-scen
Additional_Problems# planutils run val -- Validate ../Problem_2/domain.pddl problem_scaleup.pddl lama_first_
lan
Checking plan: lama_first_sas_plan
Plan executed successfully - checking goal
Plan valid
Final value: 51

Successful plans:
Value: 51
  lama_first_sas_plan 51

```

Figure 16: In order: Downward, validation of plans

8.7.2 Pod Capacity Scaled

```

move-via-tunnel-to-beta curator1 hall-beta tunnel (1)
move-via-tunnel curator1 tunnel stasis-lab (1)
unload-from-pod curator1 art-beta-3 pod1 stasis-lab slot-c1a (1)
[t=0.054409s, 12924 KB] Plan length: 54 step(s).
[t=0.054409s, 12924 KB] Plan cost: 54
[t=0.054409s, 12924 KB] Expanded 252 state(s).
[t=0.054409s, 12924 KB] Reopened 0 state(s).
[t=0.054409s, 12924 KB] Evaluated 253 state(s).
[t=0.054409s, 12924 KB] Evaluations: 506
[t=0.054409s, 12924 KB] Generated 3728 state(s).
[t=0.054409s, 12924 KB] Dead ends: 0 state(s).
[t=0.054409s, 12924 KB] Number of registered states: 253
[t=0.054409s, 12924 KB] Int hash set load factor: 253/256 = 0.988281
[t=0.054409s, 12924 KB] Int hash set resizes: 8
[t=0.054409s, 12924 KB] Search time: 0.007884s
[t=0.054409s, 12924 KB] Total time: 0.054409s
Solution found.
Peak memory: 12924 KB
Remove intermediate file output.sas
search exit code: 0

INFO      Planner time: 0.54s

54: SECURE-IN-POD CURATOR1 ART-BETA-2 POD2 HALL-BETA SLOT-C1A
55: MOVE-VIA-TUNNEL-TO-BETA CURATOR1 HALL-BETA TUNNEL
56: MOVE-VIA-TUNNEL CURATOR1 TUNNEL STASIS-LAB
57: UNLOAD-FROM-POD CURATOR1 ART-BETA-2 POD2 STASIS-LAB SLOT-C1A

time spent: 0.00 seconds instantiating 1108 easy, 0 hard action templates
0.00 seconds reachability analysis, yielding 155 facts and 624 actions
0.00 seconds creating final representation with 149 relevant facts
0.00 seconds building connectivity graph
0.00 seconds searching, evaluating 244 states, to a max depth of 3
0.00 seconds total time

[t=46.966387s, 26716 KB] f = 37, 240986 evaluated, 39898 expanded, 8 reopened
[t=46.969388s, 26716 KB] New best heuristic value for lmcut: 17
[t=46.969388s, 26716 KB] g=20, 241006 evaluated, 39900 expanded, 8 reopened
[t=192.067693s, 67372 KB] f = 38, 1001569 evaluated, 178934 expanded, 599 reopened
[t=192.069693s, 67372 KB] New best heuristic value for lmcut: 16
[t=192.069693s, 67372 KB] g=22, 1001583 evaluated, 178936 expanded, 599 reopened
[t=196.505480s, 67372 KB] New best heuristic value for lmcut: 15
[t=196.505480s, 67372 KB] g=23, 1030670 evaluated, 183656 expanded, 599 reopened
[t=196.509480s, 67372 KB] New best heuristic value for lmcut: 14
[t=196.509480s, 67372 KB] g=24, 1030697 evaluated, 183658 expanded, 599 reopened
Peak memory: 86252 KB
caught signal 24 -- exiting
Time limit has been reached.
Remove intermediate file output.sas
search exit code: 23

Driver aborting after search
INFO      Planner time: 299.42s

```

Figure 17: In order: Lama, FF, Downward

References

- [1] URL: <https://github.com/jan-dolejsi/pddl-ganttt>.
- [2] URL: <https://github.com/Allelallecie/autonomous-preservation-robots-in-interplanetary-museum-vault-scenario>.git.
- [3] "Following the university guidelines, LLMs were used as an additional and assistive tool for map topology image generation, improving the report's language quality and code implementation."